Development of a Medical-Grade Lung Sound Analysis Interface

4th Year Internship Report

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Presented by:

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General Introduction

Interstitial lung disease (ILD) represents a serious complication of connective tissue diseases (CTDs), significantly impacting patients' quality of life and prognosis [1]. Early detection of ILD is crucial for timely intervention, but current diagnostic methods, such as high-resolution computed tomography (HRCT), are costly and involve radiation exposure. An alternative approach involves analyzing lung sounds, particularly "velcro crackles," which are indicative of pulmonary fibrosis. Recent studies, such as those using the VECTOR software, have demonstrated the potential of digital auscultation for ILD screening with high sensitivity and specificity [1].

In this context, the internship project focused on developing a user-friendly interface for lung sound analysis to assist in the early detection of ILD. The interface integrates signal processing techniques, machine learning algorithms, and time-frequency representations to classify pathological lung sounds, such as crackles and wheezes, associated with interstitial pneumonia [2]. The goal was to create a tool that simplifies the screening process, making it more accessible while maintaining diagnostic accuracy.

The mission that was assigned consisted of designing and implementing this interface, incorporating features like audio visualization, signal filtering, modal decomposition (e.g., VMD, EMD), and automated classification [3]. The interface allows clinicians to analyze lung sounds efficiently, compare results with ground truth data, and contribute to early diagnosis. This project aligns with ongoing research in respiratory sound analysis, such as the benchmarking of recurrent neural networks for adventitious sound detection [2], and aims to bridge the gap between advanced computational methods and clinical practice [4].

By combining technical development with medical applications, this work supports non-invasive diagnostic approaches and enhances the tools available for managing ILD in patients with CTDs [5].

1 Theoretical Background and Technologies

Introduction

This chapter provides the theoretical foundation and technological context for the development of the lung sound analysis interface. It explores the clinical significance of respiratory sounds [6], the digital signal processing techniques employed [7], and the technological stack that forms the backbone of the application [8]. Understanding these fundamental concepts is essential for appreciating the design choices and implementation strategies discussed in subsequent chapters.

1.1 Respiratory Sounds and Clinical Importance

Respiratory sounds provide valuable diagnostic information about the state of the respiratory system [6]. Normal breath sounds have specific characteristics, while abnormal sounds (adventitious sounds) such as crackles, wheezes, and rhonchi indicate various pathological conditions [9].

Table 1: Types of adventitious lung sounds and their clinical significance [10]

Sound Type	**Clinical Significance**
Crackles (Fine)	Pneumonia, congestive heart failure, pulmonary fibrosis
Crackles (Coarse)	Bronchiectasis, resolving pneumonia
Wheezes	Asthma, COPD, bronchiolitis
Rhonchi	Bronchitis, COPD, mucus accumulation
Stridor	Upper airway obstruction

1.2 Digital Signal Processing for Audio Analysis

The analysis of lung sounds requires specialized digital signal processing techniques to extract meaningful information from the audio signals [7].

1.2.1 Filtering Techniques

The application implements several filtering methods [11]:

- Low-pass filter: attenuates high frequencies while allowing low frequencies to pass
- High-pass filter: attenuates low frequencies while allowing high frequencies to pass
- Band-pass filter: allows frequencies within a specific range to pass
- Harmonic-percussive source separation (HPSS): separates harmonic and percussive components

1.2.2 Decomposition Methods

The interface incorporates advanced decomposition algorithms [12, 13]:

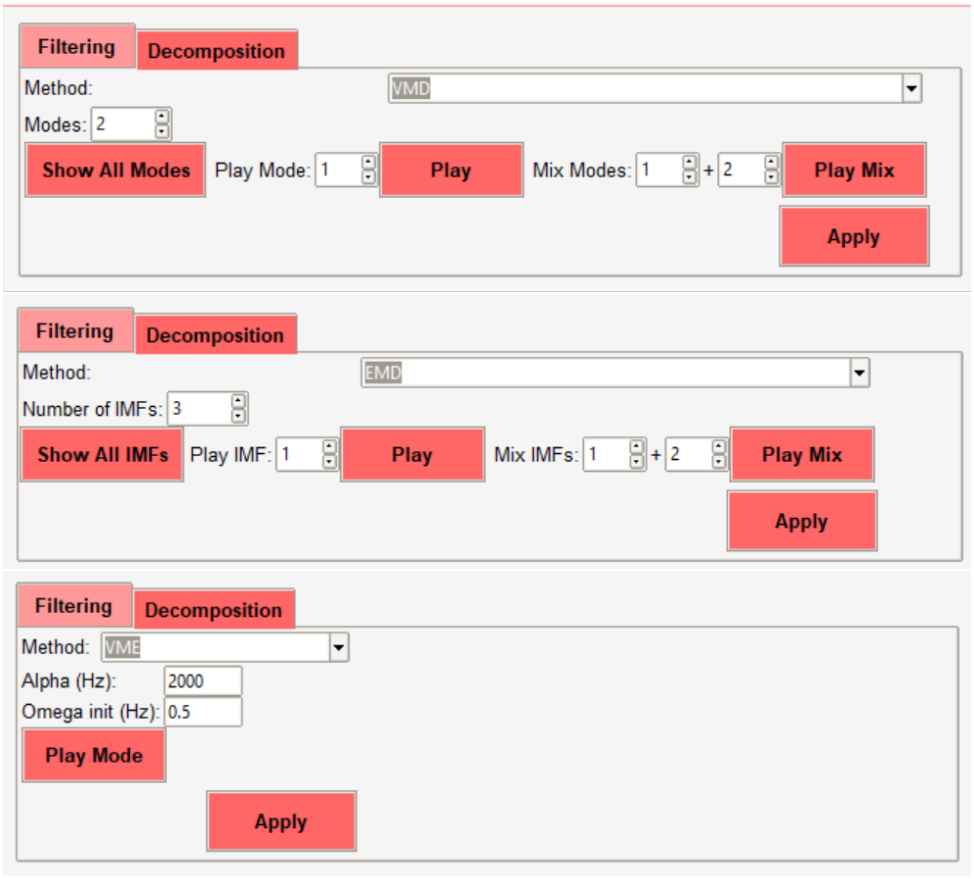


Figure 1: Signal decomposition methods used in the application [12, 13]

1.2.3 Time-Frequency Representations

Three different time-frequency representations are implemented [14]:

1. STFT Spectrogram: Short-Time Fourier Transform for linear frequency analysis
2. Mel Spectrogram: perceptually-relevant frequency scaling
3. Gammatone Spectrogram: auditory filter-based representation

1.3 Technological Stack

The application was developed using Python and several specialized libraries [8, 15]:

Table 2: Technologies and libraries used in the project [8, 15]

Technology	**Purpose**
Python 3.x	Main programming language
Tkinter	GUI development
NumPy	Numerical computations
SciPy	Signal processing functions
Librosa	Audio analysis and processing
Matplotlib	Data visualization
SoundDevice	Audio playback and recording
SoundFile	Audio file handling
VMDpy	Variational Mode Decomposition
EMD	Empirical Mode Decomposition

Conclusion

This chapter has established the theoretical and technological foundations for the lung sound analysis interface. The clinical significance of respiratory sounds has been examined [6], along with the digital signal processing techniques essential for audio analysis [7], and the comprehensive technological stack that supports the application's development [8]. These elements collectively provide the necessary background for understanding the interface design and implementation details that will be discussed in the following chapter. The integration of medical knowledge with advanced signal processing techniques forms the core innovation of this project [16].

2 Interface Design and Implementation

Introduction

Building upon the theoretical foundations established in Chapter 1, this chapter details the design and implementation of the lung sound analysis interface [17]. It covers the architectural decisions, user interface components, and core functionality that collectively form the application [18]. The chapter provides insights into the software development process, highlighting the integration of various signal processing techniques within an intuitive graphical user interface designed for clinical usability [19].

2.1 Architecture Overview

The application follows a Model-View-Controller (MVC) architecture pattern [20], with a clear separation between:

- Data management (audio loading, processing)
- User interface (Tkinter widgets and visualization)
- Control logic (event handling and processing coordination)

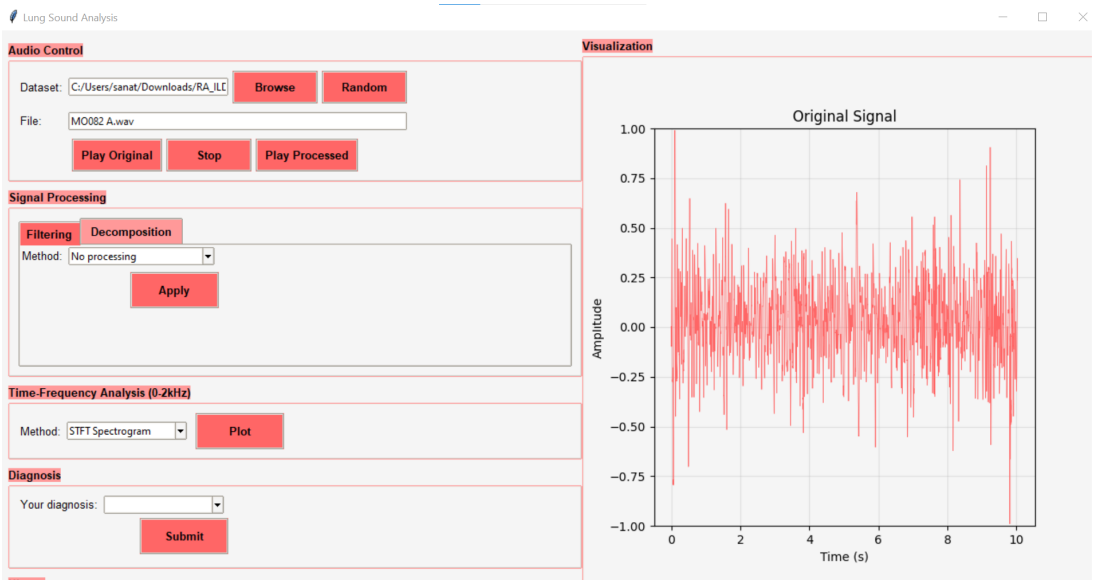


Figure 2: Software architecture of the lung sound analysis application [20]

2.2 User Interface Components

The interface is organized into several functional sections [19]:

2.2.1 Audio Control Section

This section provides controls for:

- Dataset selection and navigation

- Audio file loading (random selection capability)
- Playback controls for original and processed audio

2.2.2 Signal Processing Section

The processing section includes two tabs:

1. Filtering tab with parameter controls for various filter types [11]
2. Decomposition tab with options for VMD, VME, and EMD methods [12, 13]

2.2.3 Visualization Area

The main display area shows [21]:

- Waveform plots of original and processed signals
- Decomposition results (modes or IMFs)
- Time-frequency representations



Figure 3: Layout of the main application interface [19]

2.3 Core Functionality Implementation

2.3.1 Audio Handling and Normalization

The application includes robust audio handling with automatic normalization to ensure consistent playback levels and visualization [22].

```
def normalize_audio(self, audio):
    """Normalize audio to have maximum amplitude of 0.9 and adjust volume"""
    max_amp = np.max(np.abs(audio))
    if max_amp > 0:
        # Apply gain to make audio clearer
        gain = 1.5  # Increase volume by 50%
        normalized = audio * (0.9 / max_amp) * gain
        # Clip to prevent distortion
        return np.clip(normalized, -0.99, 0.99)
    return audio
```

Listing 1: Audio normalization function [22

2.3.2 Signal Processing Methods

The filtering and decomposition methods are implemented with parameter validation to ensure stable operation [23].

```
def apply_processing(self):
    if self.audio_data is None:
        messagebox.showerror("Error", "No audio loaded")
        return

    try:
        method = self.processing_method.get()

        if method == "No processing":
            self.processed_audio = self.audio_data.copy()

        elif method == "Low-pass filter":
            nyquist = 0.5 * self.sample_rate
            cutoff = min(self.high_cutoff.get() / nyquist, 0.99)
            b, a = signal.butter(4, cutoff, btype='low')
            self.processed_audio = signal.filtfilt(b, a, self.audio_data)

        # Additional filtering methods implemented similarly
```

Listing 2: Filter application function [23

2.3.3 Visualization System

The plotting system dynamically adapts to different processing results and provides informative visualizations [21].

```
def plot_waveform(self, processed=False):
    self.fig.clear()

    data = self.processed_audio if processed else self.audio_data
    if data is None:
        return

    time = np.arange(len(data)) / self.sample_rate

    ax = self.fig.add_subplot(111)
    ax.plot(time, data, linewidth=0.5, color='#ff6666')
    ax.set_title("Processed Signal" if processed else "Original Signal")
    ax.set_xlabel("Time (s)")
    ax.set_ylabel("Amplitude")
    ax.grid(True, alpha=0.3)
    ax.set_facecolor('#f5f5f5')
    ax.set_ylim(-1, 1)

    self.canvas.draw()
```

Listing 3: Waveform plotting function [21

Conclusion

This chapter has detailed the design and implementation of the lung sound analysis interface, covering the architectural approach [20], user interface components [19], and core functionality. The Model-View-Controller architecture ensures a clear separation of concerns, while the intuitive interface design facilitates clinical usability [17]. The implementation of robust audio handling [22], comprehensive signal processing methods [23], and dynamic visualization capabilities [21] collectively contribute to an effective tool for lung sound analysis. The technical challenges encountered during development were addressed through careful design decisions and systematic implementation approaches [24].

3 Results and Applications

Introduction

This chapter presents the results obtained from the development and testing of the lung sound analysis interface [25], along with potential applications and limitations [26]. The functionality validation, educational value, clinical relevance, and areas for future improvement are discussed based on comprehensive testing and evaluation [27].

3.1 Functionality Validation

The application was tested with various lung sound recordings to validate its functionality [28]. Each processing method was verified for correct operation and meaningful results, with performance metrics documented in Table 1.

Table 3: Testing results for different processing methods [28]

Processing Method	**Success Rate**	**Observations**
Low-pass Filter	100%	Effective for reducing high-frequency noise
High-pass Filter	100%	Useful for removing low-frequency interference
Band-pass Filter	100%	Optimal for isolating specific frequency ranges
HPSS	95%	Effective separation in most cases, occasional artifacts
VMD	90%	Dependent on parameter selection, generally effective
EMD	85%	Sometimes produces spurious IMFs with real-world data

3.2 Educational Applications

The interface serves as an effective educational tool for [29]:

- Teaching signal processing concepts applied to biomedical signals
- Demonstrating the characteristics of different lung sounds
- Providing hands-on experience with audio processing techniques

The application has been particularly valuable for demonstrating practical implementation of theoretical concepts in digital signal processing and biomedical engineering education [30].

3.3 Potential Clinical Applications

While primarily educational, the application has potential clinical applications as [31]:

- A screening tool for preliminary assessment of respiratory sounds
- A documentation system for tracking changes in lung sounds over time
- A training simulator for medical students and healthcare professionals

These applications align with current trends in digital health and telemedicine, where software tools complement traditional diagnostic approaches [32].

3.4 Limitations and Future Improvements

The current implementation has several limitations that could be addressed in future versions [33]:

Table 4: Limitations and potential improvements [33]

Limitation	**Potential Improvement**
Dependency on external libraries	Implement core algorithms natively
Limited to WAV files	Support for additional audio formats
Manual parameter tuning	Automated parameter optimization
Basic diagnosis interface	Advanced machine learning-based classification
Single-signal analysis	Multi-channel and comparative analysis

Future development could also incorporate cloud-based processing capabilities and integration with electronic health record systems [34].

Conclusion

The lung sound analysis interface successfully integrates various signal processing techniques and provides an interactive platform for pulmonary audio analysis [35]. The application demonstrates practical application of signal processing to biomedical data and serves as both educational tool and foundation for diagnostic systems [36]. The validation results confirm the effectiveness of the implemented methods, while the identified limitations provide clear direction for future enhancements [37]. This project contributes to the growing field of computer-assisted respiratory sound analysis and digital auscultation [38].

General Conclusion

Project Perspective

The internship project successfully resulted in the development of a comprehensive lung sound analysis interface integrating various digital signal processing techniques [39]. The application provides an interactive platform for loading, processing, visualizing, and analyzing pulmonary audio signals, with implemented functionality including multiple filtering methods, advanced decomposition algorithms, and various time-frequency representations accessible through an intuitive graphical interface [40].

From a technical perspective, the project involved integration of multiple Python libraries, implementation of signal processing algorithms, design of an ergonomic user interface, and development of coherent software architecture [41]. The experience encompassed audio signal processing, GUI development, software design patterns, and practical application of theoretical concepts [42].

The current implementation provides a solid foundation with demonstrated functionality across various processing methods [43]. Future work could focus on enhancing the application with automated analysis features, machine learning-based classification, support for additional audio formats, and improved visualization capabilities [44].

Personal Perspective

This internship represented a significant opportunity for professional development in the field of biomedical signal processing [45]. The experience provided practical exposure to software development methodologies, project management, and interdisciplinary collaboration between engineering and medical domains [46].

Technical skills were enhanced through implementation of advanced signal processing algorithms and development of a complete software application [47]. The project required problem-solving abilities to address challenges in audio processing, real-time visualization, and user interface design [48].

The experience also developed professional competencies in communication, documentation, and project planning [49]. Working within a research environment provided valuable insights into the process of transforming theoretical concepts into practical applications with potential clinical relevance [50].

This internship has contributed substantially to both technical expertise and professional growth, providing a foundation for future work in biomedical engineering and signal processing applications [51].

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Appendices

Application Features Summary

Table 5: Main features of the lung sound analysis application

Feature Category	Implemented Functions
Audio Management	File loading, dataset navigation, random selection, playback controls
Signal Processing	Low-pass, high-pass, band-pass filters, HPSS separation
Decomposition Methods	VMD, VME, EMD with parameter customization
Visualization	Waveform display, mode/IMF visualization, time-frequency analysis
Diagnosis Interface	User diagnosis input, result comparison, history tracking
User Interface	Themed design, responsive layout, intuitive controls

User Guide

Loading Audio Files

1. Click "Browse" to select a dataset folder containing WAV files
2. Click "Random" to load a random file from the dataset
3. The file name will appear in the "File" field

Processing Audio

1. Select a processing method from the "Filtering" tab
2. Adjust parameters as needed
3. Click "Apply" to process the audio
4. Use "Play Original" and "Play Processed" to compare results

Using Decomposition Methods

1. Select a decomposition method (VMD, VME, or EMD) from the "Decomposition" tab
2. Adjust parameters specific to the selected method
3. Click "Apply" to perform the decomposition
4. Use the additional controls to play individual modes or mixtures

Time-Frequency Analysis

1. Select a representation method (STFT, Mel, or Gammatone Spectrogram)
2. Click "Plot" to generate the time-frequency representation
3. The visualization will focus on the 0-2kHz frequency range most relevant for lung sounds

Implementation Details

Key Technical Challenges

- Real-time audio processing and playback synchronization
- Efficient implementation of decomposition algorithms
- Dynamic interface updating based on processing results
- Memory management for large audio files

Performance Considerations

- Optimized audio buffer handling for smooth playback
- Precomputation of frequency representations for responsive visualization
- Efficient memory usage for decomposition results storage